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← Placement, section 001, Summer 2 2026

PLAYGROUND 4 - Pooled questions (Test)

INSTRUCTOR

Alberto Gandolfi

New York University Abu Dhabi, United Arab Emirate

Due Date: SAT, JUN 13, 2026 4:00 PM GMT+4

REQUEST EXTENSION

Current Score: - / 100 POINTS | 0.0 %

Scoring and Assignment Information



QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
POINTS	- / 6	- / 6	- / 6	- / 6	- / 6	- / 6	- / 6	- / 6	- / 5	- / 5	- / 6	- / 6	- / 6	- / 6	- / 6	- / 6	- / 6

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.1.031.

Perform the indicated operations and simplify. Express your answer as a single fraction.

(a) $\frac{4}{5}\left(10 - \frac{5}{4}\right)$

(b) $\left(4 + \frac{1}{3}\right)\left(1 - \frac{3}{4}\right)$

2. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.2.052.

Evaluate each expression, without using a calculator. (Simplify your answers completely.)

(a) $64^{1/3}$

(b) $(-27)^{1/3}$

(c) $-\left(\frac{1}{27}\right)^{1/3}$

3. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.3.049.

Perform the indicated operations and simplify.

$$(6x - 3)(x^2 - x + 1)$$

4. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.4.019.

Simplify the rational expression.

$$\frac{x - 9}{x^2 - 81}$$

5. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.5.100.

Find all real solutions of the equation. (Enter your answers as a comma-separated list. If there is no real solution, enter NO REAL SOLUTION.)

$$|7x| = 4$$

x =

6. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

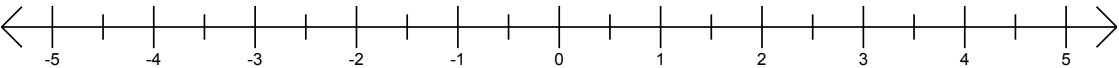
SPreCalc8M 1.8.079.

Solve the absolute-value inequality. Express the answer using interval notation.

$|3x + 5| < 7$

Graph the solution set.

Use the tools to enter your answer.



NO SOLUTION

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- -
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7. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.10.059.

Find the slope and y -intercept of the line. (If an answer does not exist, enter DNE.)

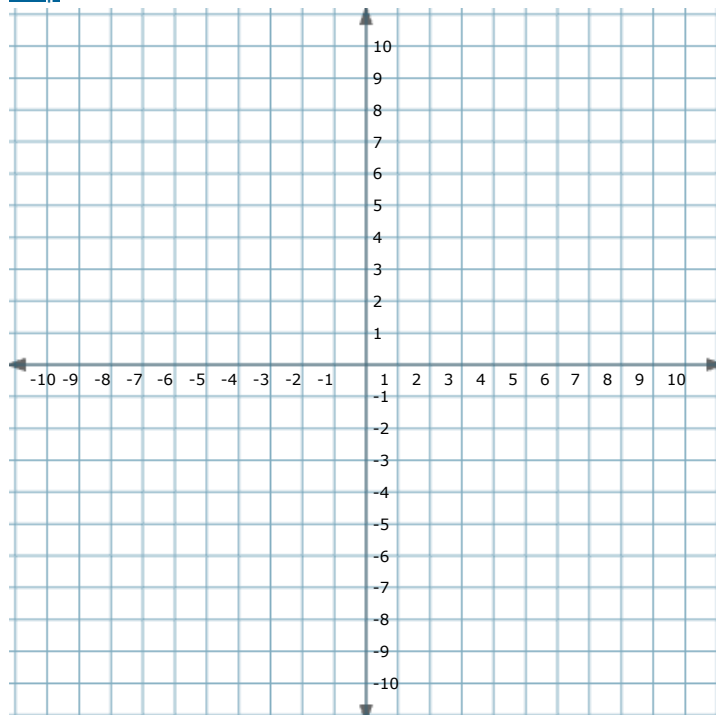
$$-4x + y = 2$$

slope

 y -intercept $(x, y) = ($ $)$

Draw its graph.

- ☐ Selection Tool
- ☐ Line
- ☐ Ray
- ☐ Segment
- ☐ Circle
- ☐ Vertical Parabola
- ☐ Horizontal Parabola
- ☐ Point

[No Solution](#)[Help](#)[Clear Graph](#)[Delete Layer](#)[Fill](#)**WebAssign Graphing Tool**

Graph LayersToggle Open/Closed

- After you add an object to the graph you can use Graph Layers to view and edit its properties.

8. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 2.1.528.XP.

Evaluate the function at the indicated values. (If an answer is undefined, enter UNDEFINED.)

$$f(x) = x^3 + 2x; \quad f(-2), f(-1), f(0), f\left(\frac{1}{2}\right)$$

$$f(-2) = \text{[input box]}$$

$$f(-2) = \text{[input box]}$$

$$f(-1) = \text{[input box]}$$

$$f(-1) = \text{[input box]}$$

$$f(0) = \text{[input box]}$$

$$f(0) = \text{[input box]}$$

$$f\left(\frac{1}{2}\right) = \text{[input box]}$$

$$f\left(\frac{1}{2}\right) = \text{[input box]}$$

9. [- / 5 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 2.2.057.

Determine whether the equation defines y as a function of x .

$$x = y^8$$

- ☐ is a function
- ☐ is not a function

10. [- / 5 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 2.3.024.

A function f is given.

$$f(x) = \sqrt{x - 6}$$

Use a graphing device to draw the graph of f . Find the domain and range of f from the graph. (Enter your answers using interval notation.)

domain

range

11. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 3.1.515.XP.

A quadratic function f is given.

$$f(x) = x^2 - 4x + 5$$

(a) Express f in vertex form.

$$f(x) = \boxed{} //$$

$$\boxed{} //$$

(b) Find the vertex and x - and y -intercepts of f . (If an answer does not exist, enter DNE.)

$$\text{vertex } (x, y) = \left(\boxed{} //, \boxed{} // \right)$$

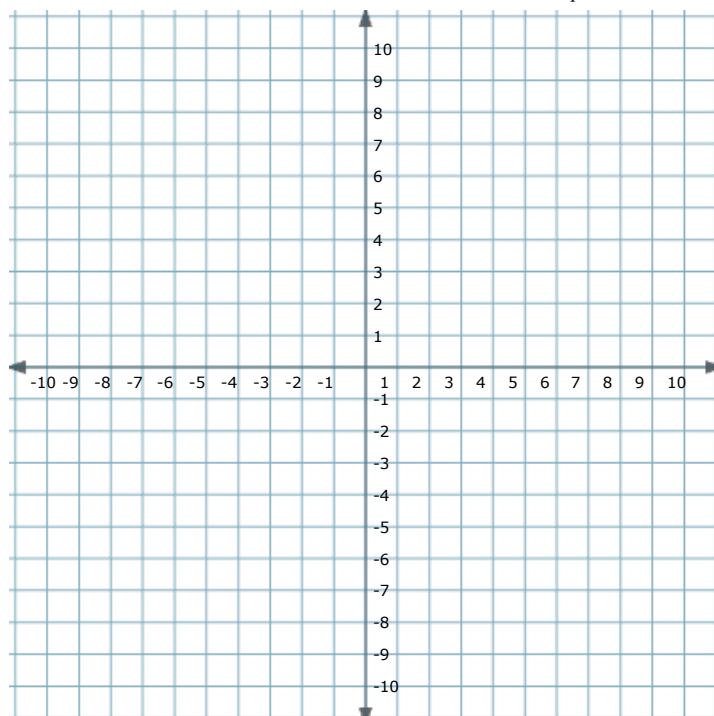
$$\text{x-intercept } (x, y) = \left(\boxed{} //, \boxed{} // \right)$$

$$\text{y-intercept } (x, y) = \left(\boxed{} //, \boxed{} // \right)$$

(c) Sketch a graph of f .

- ☐ Selection Tool
- ☐ Line
- ☐ Ray
- ☐ Segment
- ☐ Circle
- ☐ Vertical Parabola
- ☐ Horizontal Parabola
- ☐ Point

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WebAssign Graphing Tool

Graph LayersToggle Open/Closed

- After you add an object to the graph you can use Graph Layers to view and edit its properties.

(d) Find the domain and range of f . (Enter your answers using interval notation.)

domain

range

12. [- / 6 Points]

DETAILS

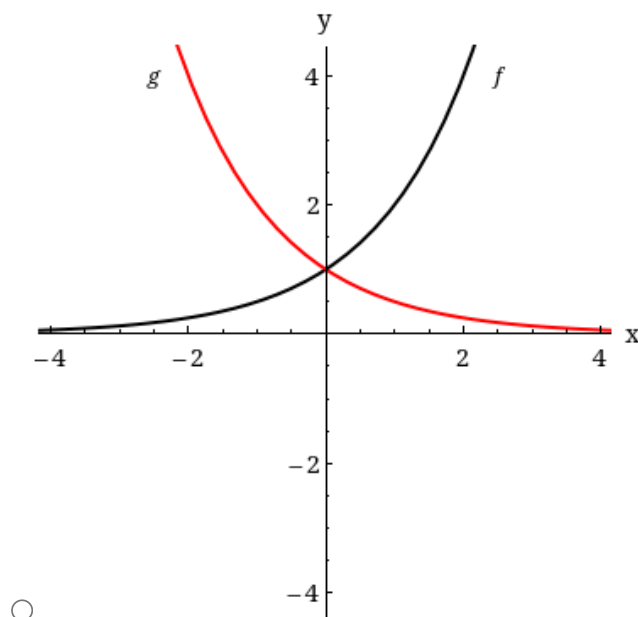
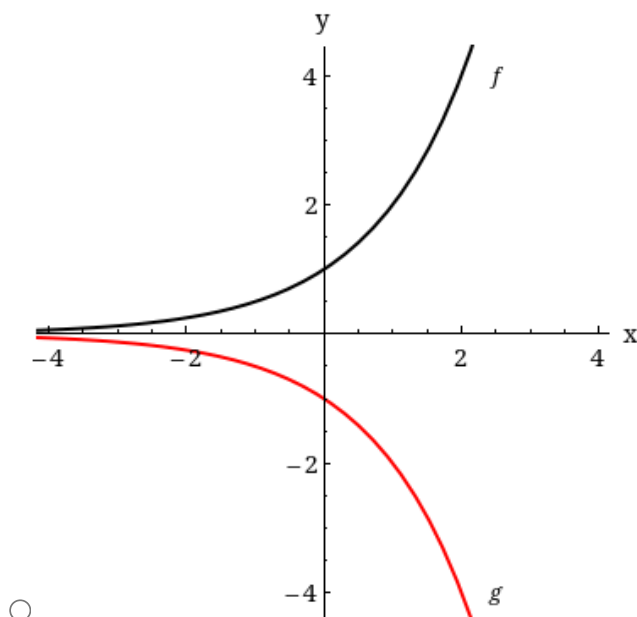
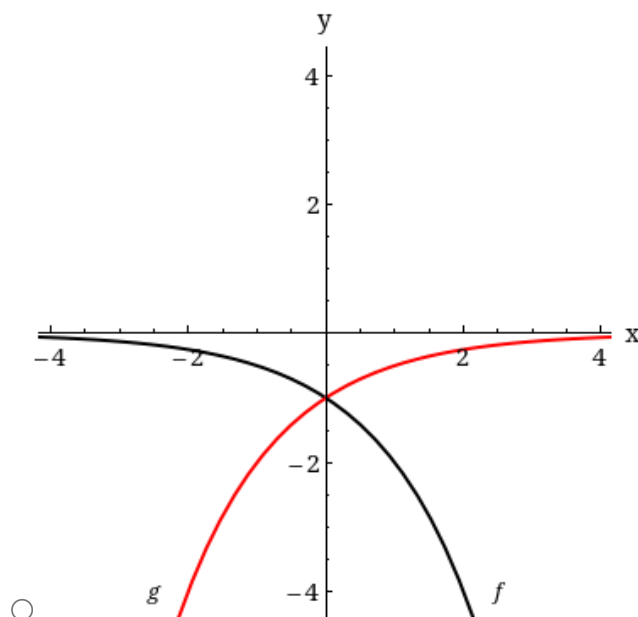
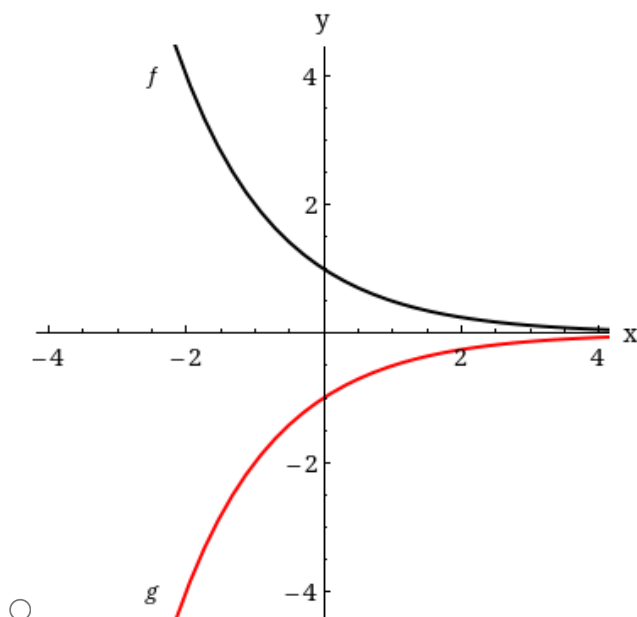
MY NOTES

ASK YOUR TEACHER

SPreCalc8M 4.1.508.XP.

Graph both functions on one set of axes.

$$f(x) = 2^x \text{ and } g(x) = 2^{-x}$$



13. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 5.1.029.

Find the terminal point $P(x, y)$ on the unit circle determined by the given value of t .

$$t = \frac{3\pi}{2}$$

$$P(x, y) = ($$

 $)$

14. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 6.1.019.

Find the degree measure of the angle with the given radian measure.

$$\frac{7\pi}{6}$$

°

15. [- / 6 Points]

DETAILS

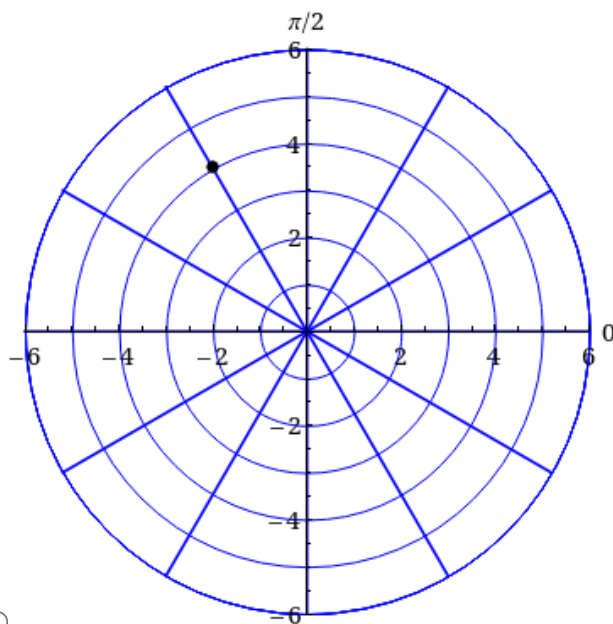
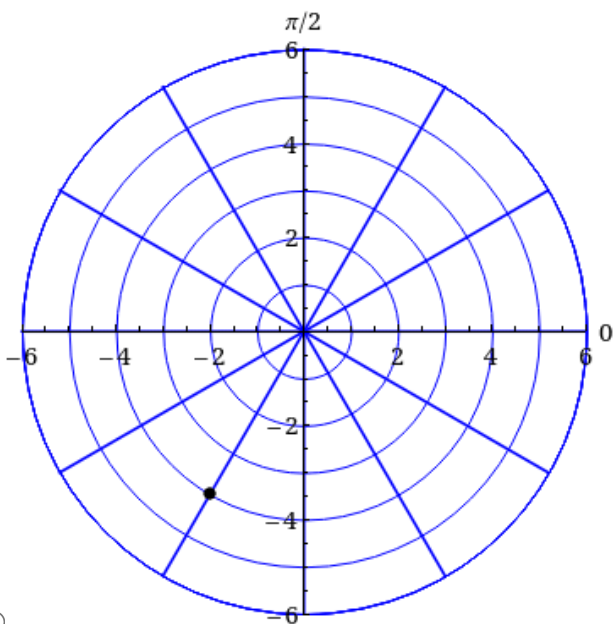
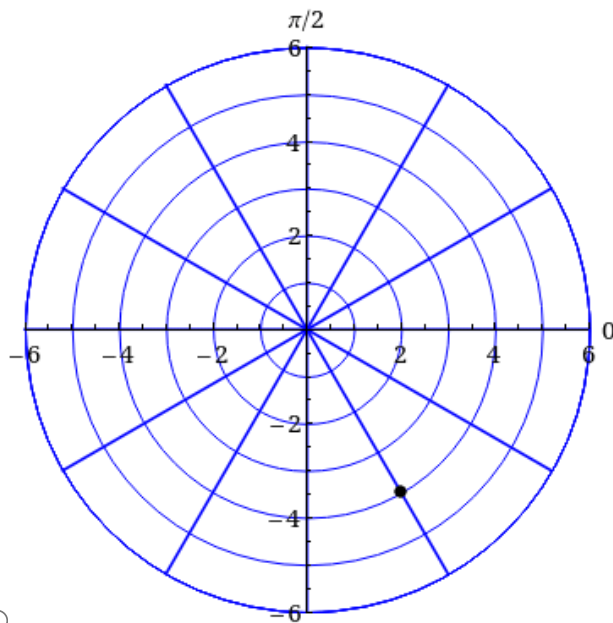
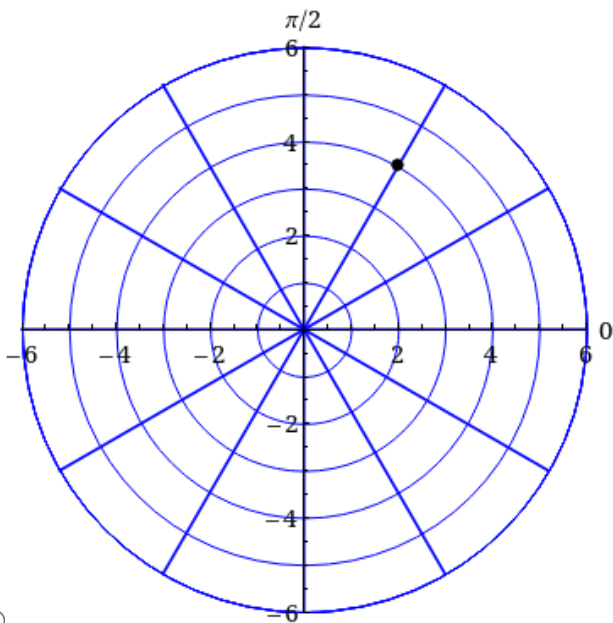
MY NOTES

ASK YOUR TEACHER

SPreCalc8M 8.1.009.

Plot the point that has the given polar coordinates.

$$\left(-4, \frac{4\pi}{3}\right)$$



16. [- / 6 Points]

DETAILS

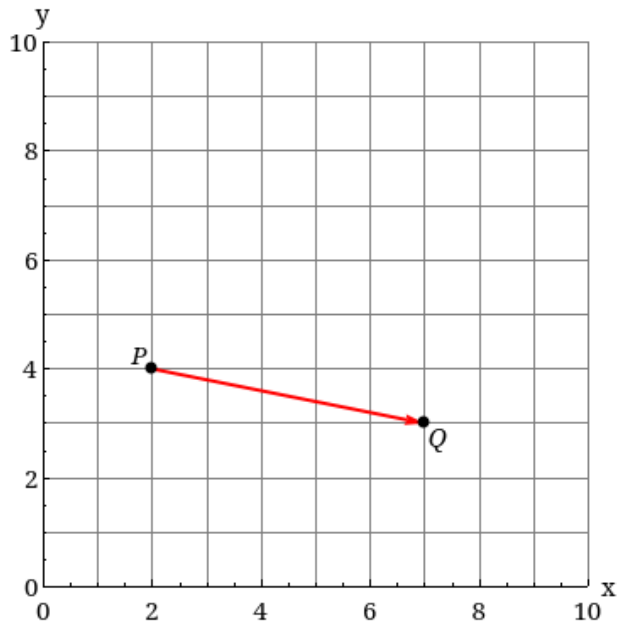
MY NOTES

ASK YOUR TEACHER

SPreCalc8M 8.5.011.

Express the vector $\mathbf{v}$ with initial point P and terminal point Q in component form. (Assume that each point lies on the gridlines.)

 $\mathbf{v} =$



17. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 11.2.019.

The first four terms of a sequence are given. Can these terms be the terms of an arithmetic sequence?

19, 12, 5, -4, . . .

☐ Yes, the sequence is arithmetic.

☐ No, the sequence is not arithmetic.

If the sequence is arithmetic, find the common difference d . (If the sequence is not arithmetic, enter DNE.)

 $d =$

